

Maulana Azad National Institute of Technology, Bhopal
Mini-Test Examination, August 2024

Department: Electrical Engineering

Semester: B. Tech. III

Subject Code: EE-214

Subject Coordinators Name: Dr. Sushma Gupta and Dr. Punjan Dohare

Date: 28th August 2024

Subject Name: Measurement and Instrumentation

Max. Marks: 25

Time: 1.00 Hrs

Set-A

Q. 1 Define and explain the following static characteristics of an Instrument:

(a) Accuracy (b) Resolution (c) Reproducibility (d) Threshold (e) Stability (5)

Q.2. Define significant figures. Two resistors R_1 and R_2 are connected in series with $R_1 = 28.7$ ohm and $R_2 = 3.624$ ohm. Calculate the total resistance to the appropriate number of significant figures. (5)

Q. 3. What is measurement. Explain significance of measurement. Define limiting error. A 0-150 V voltmeter has a guaranteed accuracy of 1 percent of full scale reading. The voltage measured by this instrument is 75 V. Calculate the limiting error in percent. (5)

Q. 4 Differentiate between indicating, integrating and recording type of instruments. (5)

Q. 5 A capacitive transducer is made up of two concentric cylindrical electrodes. The length of electrodes is 25 mm, the inner diameter of outer cylindrical electrodes is 4.2 mm and the outer diameter of inner cylindrical electrode is 4.0 mm. Determine the sensitivity of the transducer. Assume air medium. Determine the change in capacitance for a displacement of the inner electrode of 2.5 mm. Determine also the electric stress when a voltage of 150 V is applied across the electrodes. (5)